

Exact and Iterative Algorithms for Total Variation Compressed Sensing

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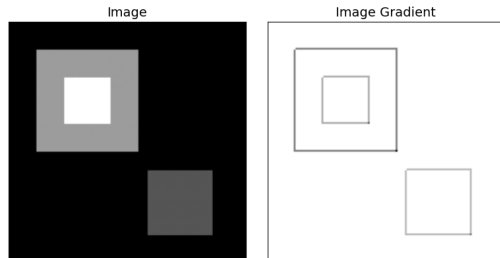
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Compressed Sensing

- ▶ Reconstructing signals and images from noisy data is very important across several applications (e.g. MRI scans)
- ▶ Classically, to recover a signal, we required *several* measurements of the signal, which was formalized by the **Shannon-Nyquist Sampling Theorem**
- ▶ In the 2000s, *Candès, Romberg, Tao*, and *Donoho* showed that a *sparse* signal can be recovered from far fewer measurements than classical sampling requires, establishing the field of **Compressed Sensing**.
- ▶ This recovery is posed as a **mathematical optimization** problem: among all signals consistent with the data, we want the simplest one

Total Variation Compressed Sensing (TVCS)

- ▶ Natural images are generally smooth, with sharp changes at their edges, which gives them a very *sparse* gradient structure
- ▶ Total Variation is used as a model for compressed sensing because it keeps edges sharp instead of blurring them out



Mathematical formulation

- ▶ We use the *anisotropic* total variation: $\text{TV}(u) = \|Du\|_1$.
- ▶ **TVCS:** $\min_u \|Du\|_1 \quad \text{s.t.} \quad Au = b$
- ▶ D is the finite difference operator, A is the partial-Fourier sampling from the limited measurements, and b is the observed Fourier data.

From 1D to 2D

1D: D is surjective, so $z = Du$ reduces TVCS *exactly* to the well-known basis pursuit problem, which is relatively simple to solve:

$$\min_z \|z\|_1 \quad \text{s.t.} \quad Mz = \tilde{b}.$$

2D: D stacks horizontal and vertical differences ($N \rightarrow 2N$), so it is *not surjective*. The original reduction underestimates the true minimum, which gives us a **relaxation gap**.

Lifted formulation (gap-free)

$$\min_{u, z} \|z\|_1 \quad \text{s.t.} \quad Au = b, \quad Du = z.$$

Coupling $Du = z$ forces $\|z\|_1 = \|Du\|_1$.

This 2D problem is the one our **algorithms** solve.

Dual Variables

- ▶ A **dual variable** p is a “price” attached to a constraint. Duality is the key tool behind both algorithms that we introduce in this talk.

Dual of our problem

$$\min_z \|z\|_1 \quad \text{s.t.} \quad Mz = \tilde{b} \quad \implies \quad \max_p \tilde{b}^\top p \quad \text{s.t.} \quad \|M^\top p\|_\infty \leq 1$$

Dual of $\min f(x)$ s.t. $Ax = b$

Lagrangian (dual variable $p \in \mathbb{R}^m$ on $Ax = b$):

$$\mathcal{L}(x, p) = f(x) + p^\top (b - Ax).$$

Dual function - find by minimizing over x :

$$g(p) = \inf_x \mathcal{L}(x, p) = p^\top b + \inf_x [f(x) - (A^\top p)^\top x] = b^\top p - \sup_x [(A^\top p)^\top x - f(x)].$$

The last supremum is the **Fenchel conjugate** $f^*(y) = \sup_x [y^\top x - f(x)]$ at $y = A^\top p$. This gives us

$$g(p) = b^\top p - f^*(A^\top p) \quad \implies \quad \max_p b^\top p - f^*(A^\top p).$$

Karush-Kuhn-Tucker (KKT) Conditions

General KKT conditions for equality-constrained problems. For a problem of the form $\min_x f(x)$ s.t. $l(x) = 0$, the Lagrangian $\mathcal{L}(x, v) = f(x) - v^T l(x)$ gives us

$$\text{stationarity: } 0 \in \partial f(x) - \sum_j v_j \nabla l_j(x) \quad \text{primal feasibility: } l(x) = 0.$$

(For an equality constraint, the Lagrange multiplier v is *free*)

For our problem, we have $f = \|\cdot\|_1$ and $l(z) = Mz - \tilde{b}$, so $\sum_j v_j \nabla l_j = M^T p$.

Stationarity gives us $0 \in \partial \|z\|_1 - M^T p \Rightarrow M^T p \in \partial \|z\|_1$, and using the subdifferential $(\partial \|z\|_1)_j = \text{sign}(z_j)$ for $z_j \neq 0$, $\in [-1, 1]$ for $z_j = 0$.

We say that pair (z^*, p^*) is optimal iff

$$\underbrace{Mz^* = \tilde{b}}_{(1) \text{ primal feasibility}} \quad \underbrace{\|M^T p^*\|_\infty \leq 1}_{(2) \text{ dual feasibility}} \quad \underbrace{(M^T p^*)_j = \text{sign}(z_j^*) \quad \forall z_j^* \neq 0}_{(3) \text{ complementarity}}$$

ADMM (Iterative)

To solve $\min_x f(x) + g(x)$, we rewrite the problem as $\min_{w,z} f(w) + g(z) \text{ s.t. } w = z$.

Algorithm 1: ADMM

Augmented Lagrangian:

$$\mathcal{L}(w, z, y) = f(w) + g(z) - \langle y, w - z \rangle + \frac{\tau}{2} \|w - z\|^2$$

for $k = 0, 1, 2, \dots$ **do**

$$\begin{array}{l} z_{k+1} = \arg \min_z \mathcal{L}(w_k, z, y_k) \\ w_{k+1} = \arg \min_w \mathcal{L}(w, z_{k+1}, y_k) \\ y_{k+1} = y_k - \tau (w_{k+1} - z_{k+1}) \end{array}$$

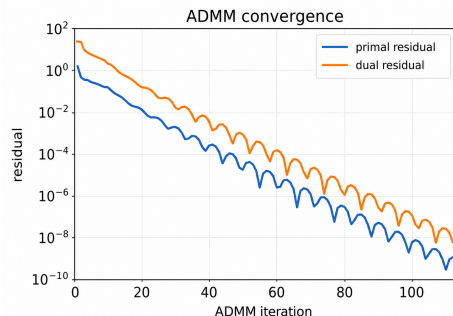
Idea: split the hard problem $f + g$ into two easier problems (w.r.t. w and z), solve each separately. Then adjust the dual variable y , until convergence.

ADMM for TVCS

To solve 2D TVCS via ADMM, we apply ADMM by splitting $Du = z$ using operator splitting methods. This gives three cheap steps:

- ▶ **Image update:** Linear / Poisson solve, which enforces $Au = b$ - solved using FFT
- ▶ **Shrinkage:** $z \leftarrow \mathcal{S}_{1/\tau}(Du + \frac{1}{\tau}y)$
- ▶ **Dual update:** $y \leftarrow y - \tau(Du - z)$, until $Du = z$

Convergence: ADMM converges linearly for TVCS.

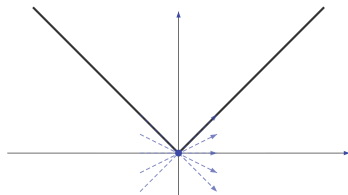


Subgradients & Differential Inclusions

When we deal with functions that aren't differentiable everywhere, we have to introduce the notion of a *subgradient*.

- ▶ For the function $|x|$, at a corner (where $x = 0$), the slope is a whole set, which is called the **subdifferential**:

$$\partial|x| = \begin{cases} \text{sign}(x), & x \neq 0 \\ [-1, 1], & x = 0. \end{cases}$$



Subgradients of the function $|x|$

Differential Inclusions

- ▶ We know that traditional gradient descent can be modeled as an ODE: $\dot{x} = -\nabla f(x)$.
- ▶ When nonsmooth, we have $\nabla f \rightarrow \partial f$, which gives a **differential inclusion**. $\dot{p} \in -\partial g(p)$, where the right hand side is set-valued.

Langlois–Darbon: The Idea

The Langlois–Darbon algorithm was initially proposed by Gabriel Langlois and Jérôme Darbon to solve the LASSO / basis pursuit problem exactly, using differential inclusions

The idea: trace the differential-inclusion trajectory (the **dual path**).

- ▶ It is **piecewise-linear**: we have straight segments, and it changes direction only at finitely many **breakpoints**.
- ▶ Between breakpoints, we move in a straight line and at each, we recompute the new direction (which is a small subproblem).
- ▶ After finitely many breakpoints, we reach the *exact* optimum.

This algorithm always gives an exact solution

The main objective of our project is to use this algorithm to build an efficient solver for 2D TVCS

Langlois–Darbon: The Algorithm

We trace the dual path from one breakpoint to the next:

Algorithm 2: Langlois–Darbon Differential Inclusion Algorithm

initialize dual variable p with $\|A^\top p\|_\infty = 1$

for $k = 1, 2, \dots$ **do**

 active set: $\mathcal{E} = \{j : |(A^\top p)_j| = 1\}$

 solve a small **NNLS** problem on $\mathcal{E} \Rightarrow$ direction d

if $d = 0$ **then return** exact solution

$\Delta \leftarrow$ step to the next **breakpoint**

$p \leftarrow p + \Delta d$

Finite termination: when $d = 0$, the exact optimum is reached.

Langlois–Darbon for TVCS

1D: TVCS reduces *exactly* to basis pursuit, and we can apply the algorithm directly.

2D: solves the lifted **partial- ℓ_1** problem (only z is penalized, the image u is *free*):

$$\min_{u, z} \|z\|_1 \quad \text{s.t.} \quad Au = b, \quad Du = z.$$

The algorithm is extended to handle the free variables.

Making the exact solver fast (our work):

- ▶ **Bottleneck:** each breakpoint re-solved a least-squares (NNLS) from scratch.
- ▶ **Warm-started NNLS:** we reused the previous active set $\Rightarrow \sim 1\text{--}2$ pivots per breakpoint, not $\sim |\mathcal{E}|$.
- ▶ **Column caching:** avoid recomputing the Poisson solves.
- ▶ **Compiled C++ core** with Matlab by using MEX, to speed up computation.

Result: 10 seconds $\rightarrow \sim 0.1$ s on a 16×16 test image.

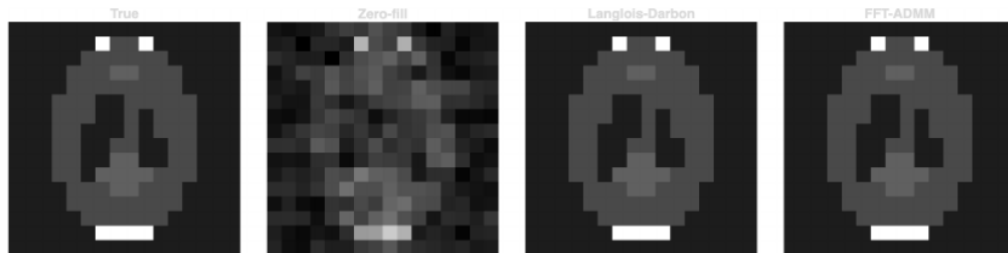
Iterative vs. Exact

	ADMM (iterative)	Langlois–Darbon (exact)
Cost per step	cheap: FFT + shrinkage	solve small NNLS on active set
Convergence	linear rate, <i>approaches</i> optimum	finite breakpoints, <i>reaches</i> optimum
Tuning	penalty parameters	none
Output	approximate (tolerance)	exact
Scales to 2D	yes	ongoing work

On 1D problems the exact solver wins outright, scaling it to be competitive in 2D and higher dimensions is the current focus of our work

Results: Reconstruction

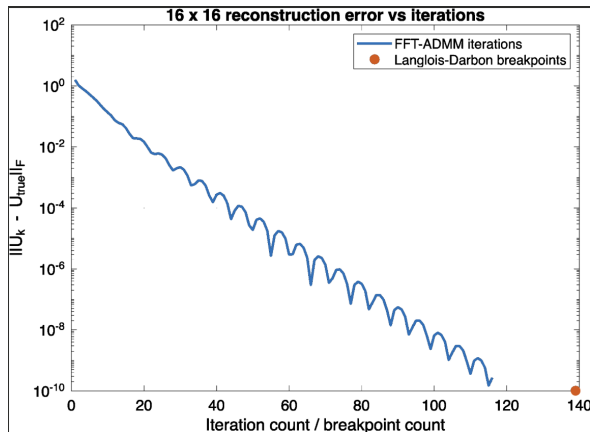
16×16 Shepp–Logan phantom recovered from **partial Fourier** measurements:



Both solvers recover the phantom. **Langlois–Darbon reaches the exact TV minimizer**, and ADMM converges to the same image.

Results: Exact v/s Iterative

Same 16×16 **Shepp–Logan** phantom reconstructed in the previous slide:



	LD	ADMM
steps	139 bp	116 it
TV gap	0	3×10^{-9}
recon error	1.3×10^{-14}	2.7×10^{-10}
time	0.28 s	0.09 s

The Langois-Darbon Algorithm hits the exact optimum, with TV gap = 0, machine precision.

ADMM is a faster, approximate algorithm.
Closing LD's wall clock time is ongoing work.

Ongoing & Future Work

- ▶ **Hybrid ADMM $\rightarrow$ LD.** Warm-start LD with ADMM's dual: a few ADMM iterations identify the support, then LD finishes and **certifies** the exact optimum in a *single* breakpoint.
- ▶ **Breakpoint conjecture.** We conjecture that the number of LD breakpoints scales with the **gradient support** $|\{j : (Du)_j \neq 0\}|$. We are currently working on a rigorous proof for this conjecture.
- ▶ **Parametric max-flow solver.** The anisotropic-TV subproblem is also a graph-cut; a Chambolle–Darbon max-flow approach could solve it exactly with a very different cost profile. We are currently building this solver, and plan to benchmark it alongside our existing 2 algorithms as well.
- ▶ **Faster Langlois–Darbon.** Optimize our implementation further to make this competitive with ADMM.

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Backup: Dual of our problem $\min \|z\|_1$ s.t. $Mz = \tilde{b}$

Here $f = \|\cdot\|_1$, so we only need f^* . It separates coordinatewise:

$$f^*(y) = \sup_z [y^\top z - \|z\|_1] = \sum_j \sup_{t \in \mathbb{R}} (y_j t - |t|).$$

Per coordinate, split on $\text{sign}(t)$:

$$t \geq 0 : (y_j - 1)t \Rightarrow \sup = \begin{cases} 0, & y_j \leq 1 \\ +\infty, & y_j > 1 \end{cases} \quad t \leq 0 : (y_j + 1)t \Rightarrow \sup = \begin{cases} 0, & y_j \geq -1 \\ +\infty, & y_j < -1 \end{cases}$$

$$\Rightarrow f^*(y) = \begin{cases} 0, & \|y\|_\infty \leq 1 \\ +\infty, & \text{otherwise} \end{cases} \quad (\text{indicator of the } \ell_\infty \text{ ball}).$$

Substituting $y = M^\top p$ into $g(p) = \tilde{b}^\top p - f^*(M^\top p)$:

$$\boxed{\max_p \tilde{b}^\top p \quad \text{s.t.} \quad \|M^\top p\|_\infty \leq 1}$$

Optimality: $M^\top p^* \in \partial\|z^*\|_1 \Rightarrow \text{supp}(z^*) \subseteq \mathcal{E} = \{j : |(M^\top p^*)_j| = 1\}$ — the LD active set.